MOBILE DRILL BRAIN

Calibration Technician Training Module 003

SPT Fundamentals, N-Value, and N60

Field	Value
Document type	Workspace 006-B technician training module
Module	003 - SPT Fundamentals, N-Value, and N60
Version	v0.2 SME-updated training draft
Prepared	July 2026
Source basis	SPT Introduction Draft v6; Chapter 2 Draft and Field Lessons; Calibration Data Collection Cheat Sheet; SME corrections supplied after Module 003 v0.1
Training lane	Calibration Technician Training - theory foundation for field setup and software workflow modules
Status	Draft for review; not a final field SOP or replacement for ASTM standards, project specifications, or manufacturer instructions

Locked module purpose
Teach a new calibration technician what SPT is, how the N-value is created, why raw N-values need energy context, why N60 exists, what ASTM standards support the work, and how to explain calibration value without giving engineering design interpretation.

1. Source basis and training boundary

This v0.2 module is grounded in the current SPT introduction draft, the hammer-energy field lessons, the calibration data-collection cheat sheet, and SME corrections supplied after Module 003 v0.1. It is intended to support the entire calibration training lane, especially the theory foundation needed before software workflow, LP/LE accuracy, refusal handling, and troubleshooting modules.

- SPT Introduction Draft v6 - SPT overview, ASTM D1586 context, N-value, N60, hammer energy, glossary, and variables that influence calibration results.
- Chapter 2 Draft and Field Lessons - hammer energy, N60 purpose, sensor installation, shallow bedrock, refusal, RPM, hammer stroke, rod connections, rod length measurement, and jobsite pace control.
- Calibration Data Collection Cheat Sheet - software workflow source for later modules, including CollectWire, Rig ID, ROD tab, SENSOR tab, top-hat check, Pause/Play, End Test, and repeated depth intervals.
- SME corrections - ASTM standards list, refusal threshold, 18-inch vs 24-inch sampler rule, depth interval clarification, N60 historical training narrative, and approved customer-facing explanation.

Training boundary
This module introduces the role of ASTM standards and N60 in SPT calibration. It does not replace the actual ASTM standards, project specifications, DOT requirements, customer requirements, analyzer manufacturer instructions, report-builder review, or engineering judgment.

2. Learning outcomes

- Explain what the Standard Penetration Test is in plain language.
- Name the basic SPT hammer and sampler setup: 140 lb hammer, 30 in. drop, 6-inch blow-count increments, and the standard N-value concept.
- Explain why the reported SPT N-value uses the second and third 6-inch increments of the drive.
- Explain why raw N-values are affected by more than soil resistance.
- Explain N60 conceptually without turning the module into a formula lesson.
- Name the training-level purpose of ASTM D1586/D1586M, D4633, D6066/D6066M, and D3740.

- State the concrete refusal threshold used in training and field practice.
- Explain the difference between 18-inch and 24-inch sampler intervals for calibration data collection.
- Use approved plain-language customer wording without giving engineering design interpretation.

3. What SPT is

The Standard Penetration Test, or SPT, is a common field test used during geotechnical drilling. A split-barrel sampler is driven into soil at the bottom of a borehole with a hammer system. Blows are recorded in 6-inch increments as the sampler advances into the soil.

The technician should understand that SPT is not just "hitting pipe." It is a standardized field test whose value depends on repeatable setup, known hammer behavior, accurate notes, and proper documentation. Calibration supports the confidence of that field data by measuring how much hammer energy is actually transferred into the rod system.

SPT element	Training-level meaning
Hammer weight	The standard SPT hammer is 140 lb unless a project-specific or special setup is documented.
Drop height	The standard SPT drop height is 30 in.
Blow increments	Blows are recorded for each 6 in. of sampler travel.
Seating drive	The first 6 in. seats the sampler and is not part of the reported N-value.
Reported N-value	The total blows for the second and third 6-inch increments.

4. ASTM standards in plain language

A calibration technician does not need to memorize full ASTM standards, but should know which standards support SPT, energy measurement, N60 context, and agency credibility. These references are included for training credibility and higher-level conversations, not to replace the standards themselves.

Standard	Training-level outline
ASTM D1586/D1586M-18e1	Core standard test method for conducting the SPT and split-barrel sampling of soils. It covers the sampler, drilling method, and recording the hammer blows required to drive the sampler.
ASTM D4633-16	Standard test method for energy measurement for dynamic penetrometers. It supports measuring energy transferred from the hammer into the drill rods so blow counts can be corrected for hammer efficiency.
ASTM D6066/D6066M-24	Standard practice for determining normalized penetration resistance of sands for liquefaction-potential evaluation. It helps explain why SPT N-values may be normalized to N60 and adjusted for overburden stress.
ASTM D3740-19	Standard practice covering minimum requirements for agencies engaged in soil and rock testing or inspection for engineering design and construction. It supports the need for qualified agencies, proper procedures, documentation, and credible reporting.

5. How the N-value is created

The N-value is a field index of penetration resistance under the standard SPT setup. In the standard explanation, the split-barrel sampler is driven 18 inches total. The first 6 inches is the seating drive. The reported SPT N-value is the total blows required to drive the sampler through the second and third 6-inch increments.

Increment	Role in standard SPT explanation
0-6 in.	Seating drive. Recorded but not included in the reported N-value.
6-12 in.	Second increment. Included in the N-value.
12-18 in.	Third increment. Included in the N-value.
N-value	Blows from 6-12 in. plus blows from 12-18 in.

Sampler length and depth interval

An 18-inch sampler is the minimum length for proper SPT data collection. A 24-inch sampler is also common and is often preferred when available because it collects more total data and makes back-end report math easier. Technicians will see both in the field.

The software depth interval field must match the actual interval being sampled. The field identifies where the sample starts and how far the sampler travels into the soil. It is entered in feet.

Example interval	Meaning
20.0 to 21.5 ft	18-inch interval
20.0 to 22.0 ft	24-inch interval

Training rule
Do not confuse the ASTM-style N-value explanation with the actual sampler interval on the job. The interval entered in the software must reflect the sampler and field setup being used.

6. Why raw N-values are not the whole story

The N-value is influenced by soil resistance, but it is not purely a soil value. Hammer efficiency, rod length, borehole conditions, sampler condition, rig setup, and drilling practices can all affect the blow count. Two rigs testing similar soil can produce different raw N-values if different amounts of energy reach the drill rods.

This is the key reason calibration matters. A raw blow count is more useful when the hammer system energy transfer is known and documented.

7. Why N60 became the reference point

For decades, SPT sampling was commonly performed with rope-and-cathead hammer systems. In those systems, the hammer was operated manually, and the energy delivered into the drill rods depended on the operator, rope wraps, hammer release, rig setup, rod string, and field conditions.

That created a problem: two crews could test similar soil and get different raw N-values because the hammer systems were not delivering the same energy. Operators also get tired during repeated field work, and manual operation can vary throughout the day.

To make SPT data more comparable, the industry adopted N60. N60 means the field N-value is normalized to a reference hammer energy of 60 percent of the theoretical hammer energy.

What the technician should remember
The technician does not need to calculate N60 in the field. The technician does need to understand why calibration matters: calibration confirms how much energy the hammer is actually transferring into the rod system so engineers can interpret blow counts on a consistent basis.

8. Hammer energy and N60

N60 is the field N-value corrected to a 60 percent theoretical hammer-energy reference. Its purpose is to put SPT results from different hammers or rigs on a more comparable basis. The formula does not need to be taught in detail in this module because labs, report builders, and engineering firms handle the calculation and interpretation.

For a calibration technician, the important idea is simple: if the actual hammer energy is known, the blow counts can be understood with better context. That is why the calibration event must collect reliable, reviewable energy-transfer data and field notes.

Approved plain-language explanation
"We are checking how much hammer energy is actually getting into the drill rods so your SPT blow counts can be corrected to the N60 standard and used with confidence."

9. Refusal rule

For training purposes, refusal means the sampler does not advance a full 6 inches in 50 blows. This is a concrete industry field standard that drillers operate by.

Condition	Training response
Sampler does not move a full 6 in. in 50 blows	Treat as refusal. Do not treat the data as normal calibration data.
Refusal begins to appear likely	Preserve the 6-inch blow-count record and document field behavior.
Refusal or obstruction affects usable data	Follow the proper field decision or escalation path. Later modules cover relocation, alternate borehole, and troubleshooting options.

10. Field variables connected to SPT theory

The following variables connect the SPT/N60 theory to what a calibration technician will see in the field. Later modules teach the setup, software workflow, data-quality checks, and troubleshooting actions in detail.

Variable	Why it matters for training
Loose sensors	A leading cause of poor data. Sensors must be secure, but not overtightened.
Hammer stroke	The hammer slug should travel the full 30-inch stroke. Reduced travel can indicate maintenance issues.
Hammer operation	Engine RPM, hydraulic pressure, and hammer cycling can affect release and repeatability.
Rod connections	Joints should be clean, tight, and flush. Gaps can reduce energy transfer.
Rod length / LE / LP	Measurement accuracy affects file setup, data meaning, and report quality.
Sampler interval	18-inch and 24-inch intervals may both appear; the actual interval must be documented in feet.
Refusal / shallow bedrock	Refusal data should not be treated as normal calibration data.
Jobsite pace	The calibration tech controls the pace. Do not sacrifice data quality for production speed.

11. Technician communication boundaries

A technician should be able to explain the purpose of calibration in plain language, but must not give engineering design interpretation, promise a specific energy value before testing, or tell the customer that the hammer has passed before data is collected and reviewed.

Customer question	Safe technician response
Why are you doing this?	We are checking how much hammer energy is actually getting into the drill rods so your SPT blow counts can be corrected to the N60 standard and used with confidence.
Will my hammer pass?	We cannot say that before collecting and reviewing the data.
Is the soil good or bad?	The engineering firm interprets the soil data. Our job is to measure and document hammer energy transfer accurately.
Can you rush the test?	The calibration tech controls the pace because reliable data and proper documentation matter.

12. What Module 003 does not teach yet

Module 003 intentionally stays at the SPT/N-value/N60 foundation level. The following items belong in later modules:

- Button-by-button SPT Analyzer software workflow - Module 004.
- LP, LE, depth interval accuracy, and file naming/data storage accuracy - Module 005.

- Loose sensors, refusal, bad graph shape, RPM issues, hammer stroke, and bad data troubleshooting - Module 006.
- Report package assembly, review responsibility, and final delivery - Module 007.
- Customer communication practice and do-not-promise scenarios - Module 008.

13. Scenario practice

Scenario: A customer says, "Our driller knows the soil out here. Why does hammer calibration matter if the blow counts are already recorded?"

Trainee response should include:

- Raw blow counts are affected by soil resistance and hammer energy transfer.
- SPT uses standard field concepts, but different hammer systems can transfer different energy into the rod string.
- N60 exists so blow counts can be corrected to a common reference.
- Calibration measures the specific hammer system so the data can be interpreted with more confidence.
- The technician should not give engineering interpretation or promise a specific result before data is reviewed.

14. Knowledge check

#	Question	Answer key
1	What is SPT?	A field test where a split-barrel sampler is driven into soil and blows are recorded as the sampler advances.
2	What hammer weight and drop height are tied to the standard SPT setup?	140 lb hammer and 30 in. drop height.
3	Which 6-inch increments make up the reported SPT N-value?	The second and third 6-inch increments.
4	Why is the first 6 inches not included in the N-value?	It is the seating drive.
5	What does ASTM D1586/D1586M support?	The SPT and split-barrel sampling field procedure.
6	What does ASTM D4633 support?	Energy measurement for dynamic penetrometers / hammer energy transfer into the drill rods.
7	What does N60 do conceptually?	It normalizes the field N-value to a 60 percent hammer-energy reference for more comparable interpretation.
8	What is the field refusal threshold taught in this module?	The sampler does not advance a full 6 inches in 50 blows.
9	What is the minimum sampler length for proper data collection?	18 inches.
10	Why might a 24-inch sampler be preferred?	It collects more data and makes back-end report math easier.
11	What must the depth interval field match?	The actual sampled interval, entered in feet.
12	What should the tech say if asked whether the hammer passed before testing is complete?	Do not promise pass/fail before data is collected and reviewed.

15. Practical signoff checklist for Module 003

Signoff item	Pass standard	Trainer initials/date
Explains SPT basics	Trainee can explain the hammer weight, drop height, 6-inch increments, seating drive, and N-value concept.	
Explains ASTM context	Trainee can identify the training-level purpose of D1586/D1586M, D4633, D6066/D6066M, and D3740.	
Explains N60 conceptually	Trainee can explain why raw N-values need hammer-energy context without trying to perform engineering interpretation.	

Applies refusal rule	Trainee states that refusal means the sampler does not advance a full 6 inches in 50 blows.	
Documents interval correctly	Trainee understands that 18-inch and 24-inch sampler intervals may both appear and must be documented in feet.	
Uses safe customer wording	Trainee explains calibration value without promising results or interpreting soil design.	

16. SME-needed items and next module inputs

Resolved in v0.2:

- ASTM standards to reference in Module 003 have been identified and reduced to training-level language.
- N60 remains conceptual; no formula-heavy explanation is needed in Module 003.
- Source terminology should be used for energy terms.
- Refusal threshold has been confirmed as the sampler not advancing a full 6 inches in 50 blows.
- 18-inch sampler is the minimum for proper data collection; 24-inch sampler is common/preferred when available for more data and easier back-end reporting math.
- Depth interval field must match actual sampler travel and is entered in feet.
- Approved customer-facing explanation has been incorporated in simplified form.

Still needed for later modules:

- Module 004 - build the button-by-button SPT Analyzer workflow from the cheat sheet and SME clarification.
- Module 005 - teach LP, LE, depth interval, file naming, and field documentation accuracy in detail.
- Module 006 - teach data quality, refusal, top-hat graph problems, loose sensors, bad data, RPM, hammer stroke, rod connections, and escalation scenarios.
- Module 007 - connect field data and hard-copy 6-inch blow counts to report-builder requirements.

Revision history

Version	Status	Change
v0.1	Draft	Initial Module 003 created from SPT Introduction Draft and Chapter 2 Field Lessons.
v0.2	SME-updated draft	Added ASTM standards outline, N60 historical/reference narrative, refusal rule, sampler/depth interval clarification, approved customer explanation, updated quiz, and revised next-module inputs.